

Penile Carcinoma In Situ and Squamous Cell Carcinoma

EPIDEMIOLOGY

Squamous cell carcinoma (SCC) of the penis accounts for 0.3 percent to 0.6 percent of cancers and less than 1 percent of cancer deaths in the United States. Approximately 400 cases and 100 deaths occur annually in the United Kingdom. In the developing world, where neonatal or infantile circumcision is not commonly practiced, penile SCC represents 10 percent to 20 percent of male tumors. However, in developed countries such as Japan and Denmark—where circumcision is not routine—the incidence of penile cancer is declining and now approximates rates seen in the United States and United Kingdom. The epidemiology of penile carcinoma in situ (PCIS) remains poorly defined, likely due to underdiagnosis and underreporting.

PENILE CARCINOMA IN SITU AND SQUAMOUS CELL CARCINOMA AT A GLANCE

Erythroplasia of Queyrat, Bowen disease of the penis, and bowenoid papulosis are clinical terms used to describe conditions—typically in uncircumcised men—where histologic examination reveals penile carcinoma in situ. **Penile carcinoma in situ (PCIS)** is the preferred term over penile intraepithelial neoplasia (PIN), to avoid confusion with prostatic intraepithelial neoplasia.

ETIOLOGY AND PATHOGENESIS

Established risk factors for penile SCC include those listed below.

Risk Factors for Penile Cancer

- Uncircumcised state
- Phimosis
- Poor hygiene
- Chronic irritation and inflammation
- Scarring
- Smoking
- Lichen sclerosus
- Lichen planus
- Squamous hyperplasia
- Bowen disease of the penis
- Erythroplasia of Queyrat
- Bowenoid papulosis
- Human papillomavirus infection
- Other sexually transmitted diseases
- Multiple sexual partners
- Human immunodeficiency virus infection
- ?Psoriasis
- Photochemotherapy
- Iatrogenic immunosuppression

CLINICAL FINDINGS

Erythroplasia of Queyrat (EQ) presents as bright red, shiny patches or plaques on mucosal surfaces—the glans and prepuce—typically in uncircumcised men.

Bowen disease of the penis (BDP) manifests as red, scaly patches or plaques on keratinized penile skin. Both EQ and BDP primarily affect older men.

Bowenoid papulosis (BP) appears as multiple warty lesions rather than uniform erythematous or scaly plaques. Lesions may be pigmented in

keratinized areas and are often multiple and more inflamed in mucosal regions.

Lesions tend to be more polymorphic in appearance.